

CEH Elite — Practical Lab Roadmap

[Step-by-step free lab sequence](#) · [TryHackMe](#) · [HackTheBox](#) · [PortSwigger](#) · [VulnHub](#) · [DVWA](#)

This roadmap gives you a numbered, ordered list of free labs to complete before your CEH Practical exam. Follow the phases in order — each phase builds on the previous one. Complete Phase 1 before touching Phase 2. Every lab is 100% free. Estimated total: 80–100 hours of hands-on practice.

7 Phases	40+ Free Labs	~90 hrs Total Practice	100% Free Resources
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Phase 1

Linux & Networking Basics — Weeks 1–2 (Do this FIRST, everything depends on it)

Before any hacking tool, you must be comfortable in the Linux terminal and understand how networks communicate. These labs are fast and free.

#	Platform	Lab / Room Name — Skill Covered	Time	URL
1	TryHackMe	Linux Fundamentals Part 1, 2, 3 Kali terminal, file ops, permissions	3 hrs	tryhackme.com/module/linux-funda
2	TryHackMe	Pre-Security Path Networking, HTTP, DNS, how the web works	4 hrs	tryhackme.com/path/outline/prese
3	TryHackMe	Network Services room FTP, NFS, SMTP, MySQL enumeration	2 hrs	tryhackme.com/room/networkservic
4	Linux Journey (web)	Grasshopper + Journeyman tracks bash commands used in every CEH tool	3 hrs	linuxjourney.com

Phase 2

Scanning & Enumeration — Weeks 2–3 (Most tested in CEH Practical)

Nmap is the single most important tool in the CEH practical exam. Master every flag here before moving on.

#	Platform	Lab / Room Name — Skill Covered	Time	URL
5	TryHackMe	Nmap room All nmap scan types, flags, output formats	3 hrs	tryhackme.com/room/furthernmap
6	TryHackMe	Nmap Live Host Discovery Ping sweeps, ARP scans, host discovery	2 hrs	tryhackme.com/room/nmap01
7	TryHackMe	Network Services 2 SMB, Telnet, FTP, NFS deep enumeration	2 hrs	tryhackme.com/room/networkservic
8	TryHackMe	Enumeration room enum4linux, LDAP, SNMP, NetBIOS	2 hrs	tryhackme.com/room/enumerationpe
9	HackTheBox	Meow (Starting Point) Telnet, first real target enumeration	1 hr	app.hackthebox.com/starting-poin
10	HackTheBox	Fawn (Starting Point) FTP anonymous login enumeration	1 hr	app.hackthebox.com/starting-poin
11	HackTheBox	Dancing (Starting Point) SMB share enumeration, smbclient	1 hr	app.hackthebox.com/starting-poin

Phase 3

System Hacking & Exploitation — Weeks 3–5

Metasploit and Meterpreter are heavily tested. Do these labs in exact order — earlier labs set up the mindset for later ones.

#	Platform	Lab / Room Name — Skill Covered	Time	URL
12	TryHackMe	Metasploit Introduction msfconsole, modules, payloads, sessions	3 hrs	tryhackme.com/room/metasploitint
13	TryHackMe	Metasploit Exploitation Running exploits, setting options, Meterpreter	3 hrs	tryhackme.com/room/metasploitexp
14	TryHackMe	Blue room MS17-010 EternalBlue on Windows — classic CEH target	2 hrs	tryhackme.com/room/blue
15	TryHackMe	Ice room Windows exploitation + privilege escalation	3 hrs	tryhackme.com/room/ice
16	TryHackMe	Kenobi room SMB + ProFTPD exploit + Linux privesc	2 hrs	tryhackme.com/room/kenobi
17	HackTheBox	Legacy (Retired — Free) MS08-067 Netapi — Metasploit exploitation	2 hrs	app.hackthebox.com/machines/Lega
18	HackTheBox	Lame (Retired — Free) Samba exploit via Metasploit on Linux target	2 hrs	app.hackthebox.com/machines/Lame
19	VulnHub	Kioptrix Level 1 First full pentest cycle — scan, exploit, root	3 hrs	www.vulnhub.com/entry/kioptrix-l
20	VulnHub	Kioptrix Level 2 Web SQLi + command injection + privesc	3 hrs	www.vulnhub.com/entry/kioptrix-l

Phase 4

Password Cracking & Sniffing — Weeks 5–6

John, Hashcat, Hydra, and Wireshark are all tested in CEH practical. These labs will make them second nature.

#	Platform	Lab / Room Name — Skill Covered	Time	URL
21	TryHackMe	John the Ripper room Hash formats, wordlists, rules, cracking modes	2 hrs	tryhackme.com/room/johntheripper
22	TryHackMe	Crack the Hash room Identify + crack MD5, SHA1, NTLM, bcrypt hashes	2 hrs	tryhackme.com/room/crackthehash
23	TryHackMe	Hydra room Brute force SSH, FTP, HTTP login forms	2 hrs	tryhackme.com/room/hydra
24	TryHackMe	Wireshark: The Basics Interface, filters, packet inspection	2 hrs	tryhackme.com/room/wiresharktheb
25	TryHackMe	Wireshark: Packet Operations Display filters, following streams, finding creds	2 hrs	tryhackme.com/room/wiresharkpack
26	TryHackMe	Sniffing & Decrypting ARP poisoning, MITM traffic capture	2 hrs	tryhackme.com/room/trafficanalys

Phase 5

Web Application Hacking — Weeks 6–8 (Highest weight in CEH Practical)

Web hacking is the biggest section in the CEH Practical. PortSwigger Web Security Academy is the best free resource in the world for this. Do every lab below.

#	Platform	Lab / Room Name — Skill Covered	Time	URL
27	PortSwigger	SQL Injection — all labs (Apprentice level) Manual SQLi, UNION attacks, blind SQLi	4 hrs	portswigger.net/web-security/sql
28	PortSwigger	XSS — Reflected & Stored (Apprentice) Reflected, stored, DOM-based XSS payloads	3 hrs	portswigger.net/web-security/cro
29	PortSwigger	Authentication vulnerabilities labs Brute force, username enum, broken logic	3 hrs	portswigger.net/web-security/aut
30	PortSwigger	File upload vulnerabilities Bypassing extension filters, shell upload	2 hrs	portswigger.net/web-security/fil
31	PortSwigger	Command injection labs OS command injection, blind injection	2 hrs	portswigger.net/web-security/os-
32	TryHackMe	OWASP Top 10 — 2021 room All 10 vulns with hands-on tasks	4 hrs	tryhackme.com/room/owasptop10202
33	TryHackMe	Burp Suite: The Basics Proxy, Intercept, Repeater setup	2 hrs	tryhackme.com/room/burpsuitebasi
34	TryHackMe	Burp Suite: Other Modules Intruder, Decoder, Comparer, Scanner	2 hrs	tryhackme.com/room/burpsuiteom
35	DVWA (local)	All DVWA modules on Low security SQLi, XSS, CSRF, File Upload, Cmd Injection	4 hrs	github.com/digininja/DVWA
36	VulnHub	Mr. Robot VM Full web + Linux pentest simulation	4 hrs	www.vulnhub.com/entry/mr-robot-1

Phase 6

Steganography, Cryptography & Covering Tracks — Week 8–9

CEH practical tests OpenStego directly. Cryptography is theory-heavy but a few practicals help it stick.

#	Platform	Lab / Room Name — Skill Covered	Time	URL
37	TryHackMe	Cryptography for Beginners Symmetric, asymmetric, hashing, PKI concepts	2 hrs	tryhackme.com/room/cryptographyi
38	TryHackMe	Hashing — Crypto 101 room MD5, SHA, salting, cracking hashes	2 hrs	tryhackme.com/room/hashincrypto
39	Local Install	OpenStego — practice hiding & extracting Data hiding, watermarking — CEH iLabs tool	2 hrs	www.openstego.com
40	TryHackMe	Dissecting PE Headers room File analysis, AV evasion concepts	1 hr	tryhackme.com/room/dissectingpeh

Phase 7

Full Simulation CTFs — Weeks 9–11 (Do these LAST as final prep)

These are full pentest simulations combining everything. Each one takes 3–6 hours and mirrors the CEH Practical exam flow exactly.

#	Platform	Lab / Room Name — Skill Covered	Time	URL
41	TryHackMe	Jr Penetration Tester Path Full path — recon, exploit, web, privesc	20 hrs	tryhackme.com/path/outline/jrpen
42	TryHackMe	Pickle Rick room Linux enumeration + web + simple exploitation	1 hr	tryhackme.com/room/picklerick
43	TryHackMe	Brooklyn Nine Nine room Steganography + FTP + SSH cracking combo	1 hr	tryhackme.com/room/brooklyninnen
44	TryHackMe	RootMe room Web shell upload + SUID privilege escalation	2 hrs	tryhackme.com/room/rootme
45	HackTheBox	Blue (Retired — Free) EternalBlue — full end-to-end simulation	2 hrs	app.hackthebox.com/machines/Blue
46	VulnHub	Basic Pentesting 1 Enumeration + web + Metasploit + privesc	3 hrs	www.vulnhub.com/entry/basic-pent
47	VulnHub	Kioptrix Level 3 Web SQLi + LFI + privilege escalation chain	3 hrs	www.vulnhub.com/entry/kioptrix-l

Recommended Daily Study Schedule

Day	Morning (1 hr)	Evening (1–2 hrs)	Goal
Mon	CEH theory module (read/watch)	TryHackMe room from current phase	1 theory + 1 lab
Tue	Review yesterday's lab notes	Continue or finish same TryHackMe room	Reinforce tools
Wed	New theory module	HackTheBox or VulnHub machine	Apply to real box
Thu	Tool drill (nmap/hydra/sqlmap commands)	PortSwigger web lab (if Phase 5)	Web skills
Fri	Weak area revision	New lab from roadmap	Progress
Sat	Full 3-hr lab session (one complete machine)		Simulation

Sun	Rest or light review of cheatsheets	Optional: CTF challenge or MCQ practice	Consolidate
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CEH Practical Exam — Key Facts & Tips

Format	20 scenario-based practical tasks — browser-based iLabs environment (Kali Linux + Windows targets)
Duration	6 hours — time is NOT usually the problem. Accuracy is.
Pass mark	70% (14 out of 20 tasks correct)
Top tools tested	Nmap, Wireshark, Metasploit/Meterpreter, Hydra, John the Ripper, Hashcat, SQLmap, OpenStego, Veracrypt
Read carefully	Each task tells you WHAT to find, not always HOW. Pick the right tool for the job.
Use --help	You can run tool help flags in the exam. nmap --help, sqlmap --help, hydra --help are all allowed.
Don't overthink	If a port is open and a service is vulnerable, use the obvious Metasploit module. Keep it simple.
Note answers	Write down answers as you go — you may need to cross-reference later tasks.

Must-Know Commands Before Exam Day

Nmap SYN scan + version + OS	<code>nmap -sS -sV -O 192.168.1.10</code>
Nmap all ports aggressive	<code>nmap -A -p- 192.168.1.10</code>
SMB enumeration	<code>enum4linux -a 192.168.1.10</code>
Crack NTLM hash (John)	<code>john --format=NT hash.txt --wordlist=rockyou.txt</code>
Crack NTLM hash (Hashcat)	<code>hashcat -m 1000 hash.txt rockyou.txt</code>
Brute force SSH (Hydra)	<code>hydra -l admin -P rockyou.txt ssh://192.168.1.10</code>
SQLmap dump database	<code>sqlmap -u 'http://target/page?id=1' --dbs</code>
Metasploit EternalBlue	<code>use exploit/windows/smb/ms17_010_eternalblue</code>
Meterpreter dump hashes	<code>hashdump</code>
Meterpreter privilege esc.	<code>getsystem</code>
Wireshark POST filter	<code>http.request.method=="POST"</code>
Steghide extract	<code>steghide extract -sf image.jpg</code>
ARP poisoning (arpspoof)	<code>arpspoof -i eth0 -t victimIP gatewayIP</code>
DNS zone transfer	<code>dig axfr @nameserverIP domain.com</code>

All labs listed are free at time of writing. TryHackMe free rooms may require account signup. HackTheBox retired machines require a free account. VulnHub VMs are downloaded and run locally in VirtualBox/VMware. PortSwigger Web Security Academy is fully free with no account needed for most labs.

You can do this. Consistency beats intensity — one lab a day is enough. Good luck!